ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MATHEMATICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) fundamental subject, Level 3. Exponents, logarithms and equations (25), functions (20), financial mathematics (15), trigonometry (25), analytical geometry and measurement (15).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number the answers according to the numbering system used in this question paper.
3. Show ALL calculations, diagrams and graphs that you used to find your answers. Answers only will not necessarily earn full marks.
4. An approved scientific calculator (non-programmable and non-graphical) may be used, unless stated otherwise.
5. Round off answers to TWO decimal places, unless stated otherwise.
6. Diagrams are not necessarily drawn to scale.
7. Write neatly and legibly.

QUESTION 1: EXPONENTS, LOGARITHMS AND EQUATIONS

1.1 Simplify without using a calculator: $8^{2/3} \times 27^{-1/3}$ (3)

1.2 Without using a calculator, determine: (a) $\log_2 32$ (b) $\log 50 + \log 2$ (3)

1.3 Solve for x: $3^{x+1} = 81$ (2)

1.4 Solve for x by using the quadratic formula: $2x^2 - 5x - 4 = 0$ (4)

1.5 Use logarithms to solve for x: $2^x = 50$ (3)

1.6 Determine the nature of the roots of $x^2 - 6x + 9 = 0$ without solving the equation. (2)

1.7 Solve simultaneously: $y = x + 1$ and $y = x^2 - 5$ (5)

1.8 Solve for x: $\log x + \log(x - 3) = 1$ (3)

[25]

QUESTION 2: FUNCTIONS

2.1 Given: $f(x) = \frac{6}{x} - 2$

2.1.1 Write down the equations of the asymptotes of f. (2)

2.1.2 Calculate the x-intercept of f. (2)

2.1.3 Sketch the graph of f, showing the asymptotes and the intercept. (3)

2.1.4 Write down the domain of f. (1)

2.1.5 Calculate $f(-2)$. (2)

2.1.6 Describe the transformation from $y = \frac{6}{x}$ to f. (2)

2.2 Given: $g(x) = 2^x - 4$

2.2.1 Calculate the y-intercept of g. (1)

2.2.2 Calculate the x-intercept of g. (2)

2.2.3 Write down the equation of the asymptote and the range of g. (2)

2.2.4 Sketch the graph of g. (3)

[20]

QUESTION 3: FINANCIAL MATHEMATICS

3.1 R25 000 is invested at 9% per annum, compounded monthly, for 4 years. Calculate the final amount. (3)

3.2 Calculate the effective annual interest rate if the nominal rate is 12% per annum, compounded quarterly. (3)

3.3 A company buys a machine for R180 000. Calculate its value after 5 years if it depreciates at 15% per annum on:

3.3.1 the reducing-balance method (3)

3.3.2 the straight-line method (2)

3.4 How many years will it take for an investment to double at 8% per annum, compounded annually? (4)

[15]

QUESTION 4: TRIGONOMETRY

4.1 Without using a calculator, determine the value of: $\sin 30° \cdot \cos 60° + \tan^2 45°$ (3)

4.2 Simplify: $\dfrac{\sin(180° - x) \cdot \cos(360° - x)}{\sin(-x) \cdot \cos(180° + x)}$ (4)

4.3 Prove the identity: $\dfrac{1 - \cos^2 x}{\sin x \cdot \cos x} = \tan x$ (3)

4.4 Solve for x if $x \in [0° ; 360°]$: $2 \sin x = -1$ (3)

4.5 In ΔABC, $a = 12$ cm, $b = 9$ cm and $\hat{C} = 50°$.

4.5.1 Calculate the length of c. (3)

4.5.2 Calculate the area of ΔABC. (2)

4.5.3 Calculate the size of $\hat{B}$. (3)

4.6 Sketch the graph of $y = 2 \cos x$ for $x \in [0° ; 360°]$. Show the intercepts with the axes and the turning points. Write down the amplitude and the period. (4)

[25]

QUESTION 5: ANALYTICAL GEOMETRY AND MEASUREMENT

5.1 Given the points A(−2 ; 3) and B(4 ; −5):

5.1.1 Calculate the length of AB. (2)

5.1.2 Determine the coordinates of the midpoint M of AB. (2)

5.1.3 Calculate the gradient of AB. (2)

5.1.4 Determine the equation of the line through A and B. (2)

5.1.5 Write down the gradient of a line perpendicular to AB. (1)

5.2 A solid cone has a base radius of 3 cm and a perpendicular height of 4 cm. Use the π button on your calculator.

5.2.1 Calculate the slant height. (2)

5.2.2 Calculate the volume. ($V = \frac{1}{3}\pi r^2 h$) (2)

5.2.3 Calculate the total surface area. ($A = \pi r^2 + \pi rs$) (2)

[15]

TOTAL: 100

END OF QUESTION PAPER